

Smart Sorting – Rotten Fruit Detection using Machine Learning

1. INTRODUCTION

Project Overview:

This project focuses on detecting rotten fruits using Machine Learning and image classification techniques. The system analyzes fruit images and automatically classifies them as **fresh or rotten**.

Purpose:

To reduce food wastage, improve fruit quality monitoring, and assist farmers, vendors, and distributors by providing an automated fruit quality detection system.

2. IDEATION PHASE

Problem Statement:

Manual fruit sorting is time-consuming and sometimes inaccurate, which leads to food wastage, financial losses, and reduced customer satisfaction.

Empathy Map:

The project addresses the concerns of farmers, fruit vendors, distributors, and food quality inspectors who require efficient methods to detect spoiled fruits.

Brainstorming:

The team identified key components including:

- Image dataset collection
- Image preprocessing
- Machine learning model training
- Fruit classification (fresh vs rotten)
- Web interface for prediction

3. REQUIREMENT ANALYSIS

Customer Journey Map:

User uploads fruit image → System processes image → Model predicts fruit condition → Result displayed (Fresh / Rotten).

Solution Requirements:

- Accurate fruit classification
- Fast prediction response
- Simple user interface
- Image preprocessing and model training pipeline

Data Flow Diagram:

User → Web Interface → Backend Server → Image Preprocessing → ML Model → Prediction Result.

Technology Stack:

- Python
- Flask (Web Framework)
- TensorFlow / Keras
- OpenCV
- NumPy
- HTML/CSS (Frontend)

4. PROJECT DESIGN

Problem–Solution Fit:

The system addresses inefficient manual sorting and fruit spoilage detection using automated image-based classification.

Proposed Solution:

A Machine Learning-based system that detects rotten fruits using image classification techniques and displays the result through a web application.

Solution Architecture:

The system consists of the following layers:

- User Interface (Web application)
- Backend Server (Flask)
- Image Preprocessing Layer
- Machine Learning Model (CNN)
- Prediction Output

5. PROJECT PLANNING & SCHEDULING

The project was developed using an **Agile-based approach**, covering:

- Dataset collection and preprocessing
- Model training and evaluation
- Web application development
- Integration of model with Flask backend
- Testing and performance evaluation

6. FUNCTIONAL AND PERFORMANCE TESTING

Performance Testing:

The model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score

The system was tested to ensure **fast prediction performance (<1 second per image)**.

7. RESULTS

The system successfully classifies fruit images as **Fresh or Rotten**.

The output includes:

- Predicted fruit condition
- Confidence score of prediction

The web interface displays the prediction result clearly to the user.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Automated fruit quality detection
- Reduces food waste
- Faster than manual inspection
- Scalable for agricultural applications

Disadvantages

- Requires a large dataset for better accuracy
- Performance depends on image quality
- May require retraining for different fruit types

9. CONCLUSION

The **Smart Sorting – Rotten Fruit Detection using Machine Learning system** provides an automated and efficient solution for identifying spoiled fruits. By leveraging machine learning and image processing techniques, the system improves fruit quality monitoring and helps reduce food wastage.

10. FUTURE SCOPE

- Integration with real-time camera detection
- Deployment on mobile devices for farmers
- Integration with automated fruit sorting machines
- Detection of multiple fruit diseases

- Cloud-based agricultural monitoring system

11. APPENDIX

Source Code: Available in the project repository.

Dataset: Fruit Fresh vs Rotten dataset.

GitHub Repository:
Smart-Sorting-Rotten-Fruit-Detection

Project Demo: To be provided.